

War News Intelligence Dashboard

Problem Statement

Modern conflict data is often scattered across multiple news reports, making it difficult for analysts, researchers, journalists, and policymakers to identify patterns, monitor escalation trends, compare conflicts, and understand the humanitarian impact of war events.

The challenge was to develop an interactive analytical platform that transforms raw war-event reports into actionable insights through data visualization and exploration.

The system needed to:

- Consolidate and analyze war-related events from multiple global conflicts.
- Identify geographic hotspots of conflict activity.
- Monitor escalation and de-escalation trends over time.
- Compare the intensity and impact of different conflicts using event frequency and casualty metrics.
- Analyze the distribution of warfare tactics and event types.
- Detect high-impact incidents through fatality analysis.
- Evaluate reporting patterns and coverage biases across news organizations.
- Enable users to dynamically explore conflict data using interactive filters and visual analytics.

Solution

Developed a War News Intelligence Dashboard using Python, Dash, and Plotly that provides an interactive environment for analyzing global conflict events.

Key features include:

- Interactive geospatial conflict mapping.
- Time-series conflict trend analysis.
- Multi-conflict comparison dashboard.
- Event type distribution analytics.
- Fatality impact and casualty trend analysis.
- News source reporting pattern analysis.
- Dynamic filtering by conflict, date range, and event type using Dash callbacks.

The dashboard converts raw war-event data into a decision-support tool that helps users uncover trends, compare conflicts, and explore geopolitical developments through real-time interactive visualizations.

Technologies Used

- Python
- Dash
- Plotly

- Pandas
- Data Cleaning & Feature Engineering
- Interactive Data Visualization
- Geospatial Analytics
- Dashboard Development